

Engineering Physics (2025)

Course code 25PY101

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Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

Laser

Optical fiber

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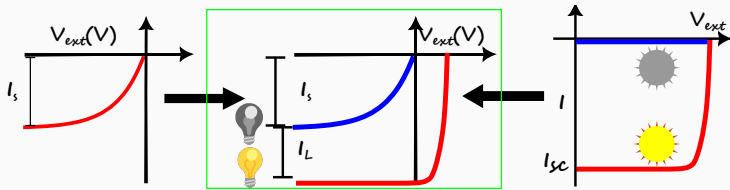
Optical fiber

Definition

A photo-diode is an opto-electronic device that is used to sense the presence of light.

- A photo-diode is a pn junction diode that operates under **reverse bias**.
- Due to the reverse bias, the depletion region width increase. This leads to decrease in junction capacitance.

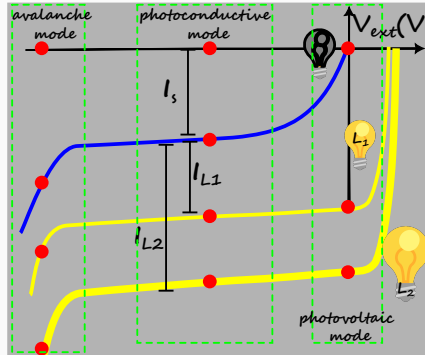
Photo-diode – $I - V$ characteristics



- The $I - V$ characteristics of photodiode can be considered as $I - V$ characteristics of solar cell under forward bias and $I - V$ characteristics of pn junction under reverse bias.
- The reverse bias generates reverse current that saturates to reverse saturation current I_s at high bias.
- In addition to the reverse current, if the photons are incident on the diode there is an additional photocurrent I_L .
- Thus the external bias is negative and the net current is negative and the $I - V$ characteristics is given by

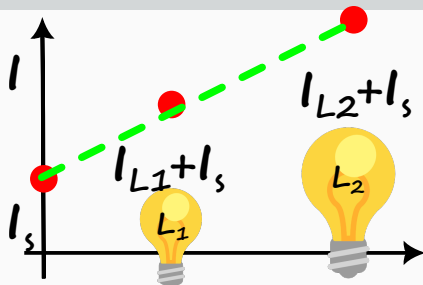
$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right] - I_L$$

Photo-diode – Modes of operation



- The photo diode has three modes of operation
 1. photoconductive mode – reverse bias
 2. zero bias mode – reverse bias voltage is set to zero
 3. avalanche mode – **very** high reverse bias
- In every mode of operation, there are two states of operation
 1. Photons are not incident 💡
 2. Photons are incident 💡

Photoconductive mode



- Photo diode is under reverse bias.
- At zero light intensity, there is a non zero current due to the reverse saturation current. This is called **dark current**.
- Response of photodiode is linear. In this mode, photodiode is a linear sensor.

- Due to reverse bias, the depletion region width increases. Since the space charge region acts as a parallel plate capacitor, the junction capacitance C_{junction} is given by

$$C_{\text{junction}} = \frac{\epsilon A}{W}$$

where A is the cross sectional area of the depletion region and W is the depletion region width.

- Due to reduction in junction capacitance, photodiode is used for high frequency opto-electronic converters.

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

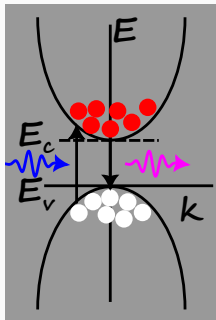
Laser

Optical fiber

Direct band gap semiconductor

Definition

A direct band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **same** value of wave vector.



- Consider electron moves from E_c to E_v .
- In the radiative recombination process,
 - Energy of electron is transferred to energy of photon.
 - Initial momentum of electron is equal to final momentum.
 - Therefore, both the energy and momentum of the (electron + hole + photon) system is **conserved**.
 - **three** particles are involved



Key Insight

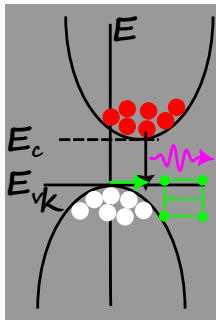
Radiative type of recombination is the dominant mechanism in a direct bandgap semiconductor.



Indirect band gap semiconductor

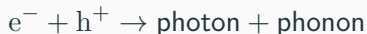
Definition

An indirect band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **different** values of wave vector.



- Consider electron moves from E_c to E_v .
- In the radiative recombination process,
 - Energy of electron is transferred to energy of photon.
 - Initial momentum of electron is not equal to final momentum.
 - Therefore, energy of the (electron + hole + photon) system is **conserved** but the momentum is not conserved.

- To conserve momentum, a **phonon assisted radiative recombination** occurs where difference in momentum is supplied by phonon and **four** particles are involved



Direct vs indirect bandgap semiconductor

SNo.	Property	Direct	Indirect
1	k at E_C vs k at E_V	equal	unequal
2	Recombination mechanism	Radiative	Non-radiative
3	Particle emitted	Photon	Phonon
4	Periodic table	Compound s.c. like GaAs, InP	Elemental s.c. like Si, Ge
5	Applications	LED, laser diode	transistors, diodes

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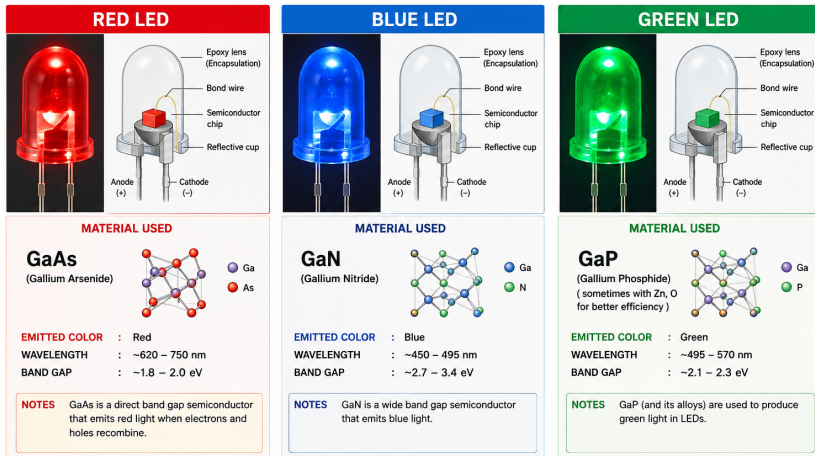
Definition

Light Emitting Diode (LED) is the opto-electronic device based on radiative recombination of charge carriers.

- LED is based on pn junction diode under forward bias.
- Under forward bias, the electrons and holes crossing the depletion region undergo radiative recombination leading to generation of light.
- From the perspective of electroluminescence, it is useful to use a semiconductor with a direct band gap semiconductor.

LED – construction

LED COLORS AND SEMICONDUCTOR MATERIALS USED

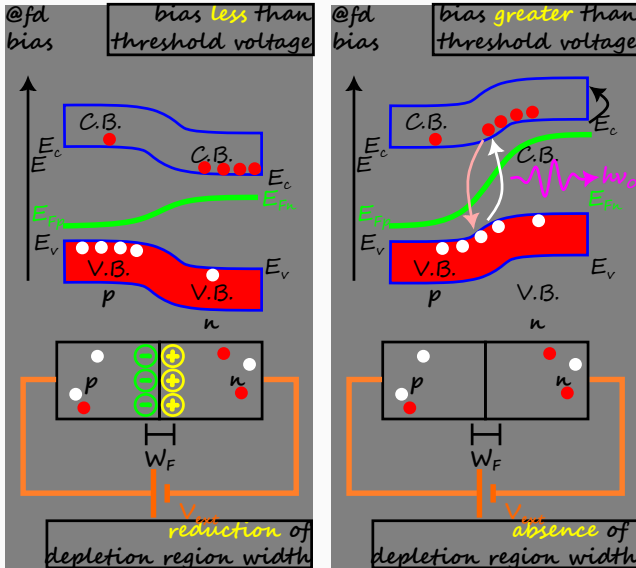


SUMMARY | LED color depends on the semiconductor material and its band gap.

Higher band gap → shorter wavelength (blue) | Lower band gap → longer wavelength (red)

- New materials AlGaInP for Red, $\text{In}_x\text{Ga}_{1-x}\text{N}$ for Green, and $\text{In}_x\text{Ga}_{1-x}\text{N}$ for Blue.

LED – working principle



- LED works on the principle of **electroluminescence**.
- Below threshold voltage, potential energy barrier is reduced, and depletion region width is reduced.
- Above threshold voltage, potential energy barrier is removed, and depletion region width is removed. The electrons and holes move towards the junction, and **recombine radiatively** to emit photon.

LED – optical emission

- From the Planck's quantum hypothesis, the wavelength can be related to the energy of photon by

$$\begin{aligned} E &= h\nu, \\ &= \frac{hc}{\lambda}, \\ E \text{ (in eV)} &= \frac{1.24}{\lambda \text{ (in } \mu\text{m)}} \end{aligned} \tag{A}$$

Problem

Prove Equation. A.

Problem

Find the energy of red photon with wavelength 700 nm.

Problem

Find the energy of blue photon with wavelength 400 nm.

- Night lighting
- Traffic signals
- Optical fibre communications
- Medical imaging

Summary of opto-electronic devices

- From the perspective of bias

Reverse bias	Zero bias	Forward bias
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Photo-diode	Solar cell	LED
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- From the perspective of depletion region width

Larger width	Nominal width	Smaller width
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Low current	Nominal current	Large current
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- From the perspective of power consumption

Low Power	Nominal Power	High Power
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Consumption	Generation	Consumption
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- From the perspective of capacitance

Low Capacitance	Nominal Capacitance	High Capacitance
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High frequency	—	—
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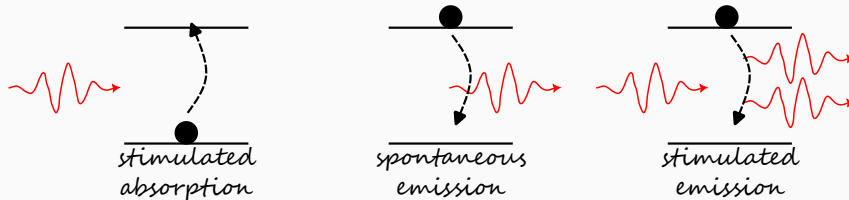
Light Emitting Diode

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Laser – Three Quantum Processes

- L.A.S.E.R is acronym for Light Amplification by Stimulated Emission of Radiation.
- Einstein predicted the possibility of lasing in 1916 by introducing concept of stimulated emission.



- Light interacts with matter by three quantum processes
 - Stimulated absorption
 - Spontaneous emission
 - Stimulated emission

Laser – Three Quantum Processes (contd...)

Definition

Stimulated absorption occurs when electron transitions from lower energy level to higher energy level upon the absorption of photon.



Definition

Spontaneous emission occurs when electron transitions from higher energy level to lower energy level by the emission of photon.



Definition

Stimulated emission occurs when electron transitions from higher energy level to lower energy level in the presence of a photon by the emission of additional photon.



Laser – Conditions and requirements for lasing action

From the Einstein's theory ¹ of light-matter interaction, the conditions for lasing action are –

- high photon density,
- large lifetime of excited levels, and
- high population of excited levels to achieve lasing action.

To satisfy these three conditions for lasing action, the requirement of lasing action are –

1. Population inversion
2. Meta-stable energy level
3. Radiation confinement

¹Refer to Chapter 44 Avadhanulu for a discussion on this theory.

Thermal equilibrium and normal state

Definition

Maxwell-Boltzmann statistics says the probability of occupying higher energy level is lower than the probability of occupying lower energy level. The probability of occupying energy level E is proportional to

$$P_{MB} \propto \exp\left(-\frac{E}{k_B T}\right)$$

Definition

Normal state is defined as the state under thermal equilibrium.

- Under the conditions of thermal equilibrium, the occupancy of energy levels is governed by Maxwell Boltzmann statistics.



Problem

Find the ratio of populations of two states in a He–Ne laser that produces light of wavelength 6328 Å at 27 °C.



normal state

To satisfy the above three requirements of lasing action, we need three components to any laser –

1. Active medium
2. Optical pump
3. Optical resonant cavity

Laser – characteristics

Laser light 	Ordinary light 
Unidirectional	Non-directional
Coherent	Incoherent
Monochromatic	Non-monochromatic
High intensity	Low intensity
Low pulse width	High pulse width

- Coherent means photons in the beam have same phase
- Monochromatic beam consists of photons of single color/wavelength.

Diode laser – Construction

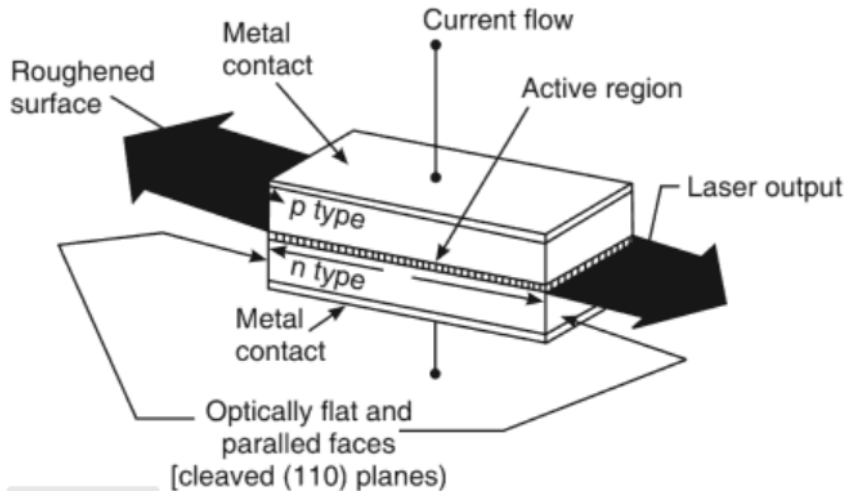
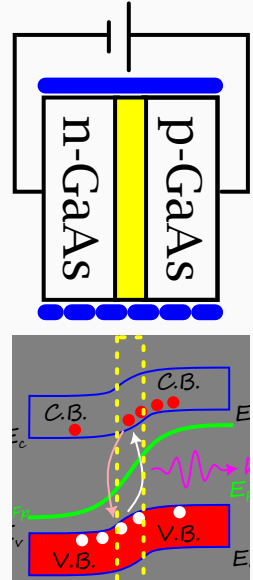


Fig. 44.24:

Schematic of homojunction diode laser

Diode laser – working principle

- In a diode laser
 - Depletion region is the active medium,
 - Forward biasing is the pump, and
 - Polished faces of the GaAs crystal are the optical resonator.
- Since the forward biasing populates the electrons in the n region and holes in the p region, when they meet at the depletion region, the active medium is already in the lasing state.
- Therefore, there is no need for a meta-stable state.



Diode laser – applications

- Industrial precision cutting
- Lasik surgery
- LiDAR (Light Imaging, Detection and Ranging) technology for distance measurement
- Optical fibre communication technology

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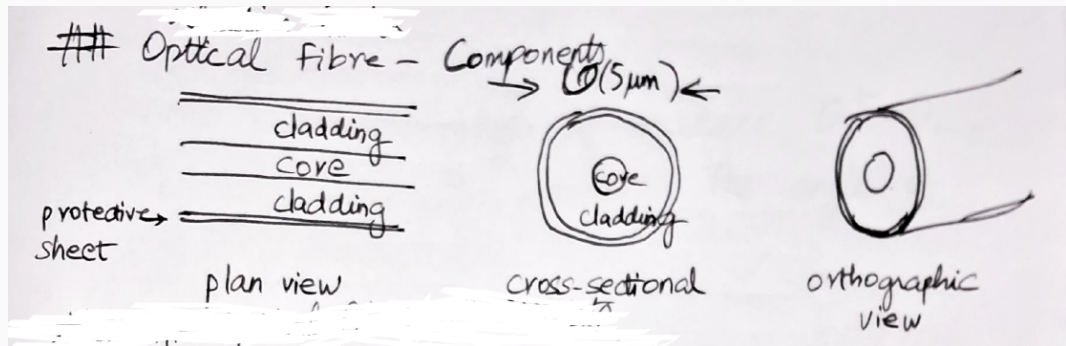
Laser

Optical fiber

Optical fiber – light pipe

- Optical fiber is an opto-electronic device that is the backbone of modern communication systems.
- Useful for long distance communication.
- Harnesses the speed of light and property of light propagation across media with different refractive indices.

Optical fiber – construction



Optical fiber – working principle

- The speed of light varies in different media.
- It is highest in vacuum and is given by $v = c$.
- In other medium, it is quantified by the refractive index n , so that the velocity of light in the medium is

$$v = \frac{c}{n}.$$

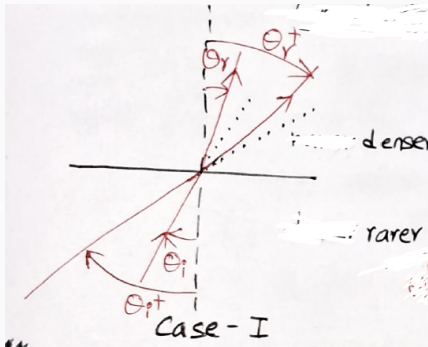
- At the interface of media, speed of light changes leading to two cases.
 - Case I: Light propagates from rarer to denser medium.
 - Case II: Light propagates from denser to rarer medium.

medium	n
air	1
water	1.33
glass	1.5

Table 1: Refractive indices of commonly available media.

Case I: Rarer to denser medium

Consider light is incident at an angle of incidence θ_i with respect to normal.

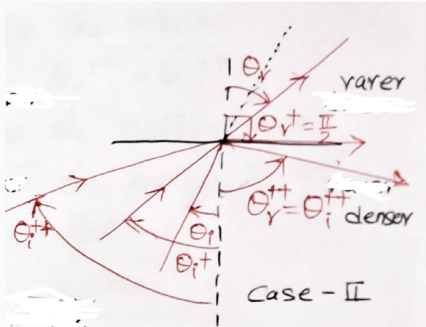


- Since light is faster in rarer medium, and slower in denser medium, it bends towards the normal. Therefore, it refracts.
- Angle of refraction is less than angle of incidence,

$$\theta_r < \theta_i.$$

- Similarly, for larger angle of incidence θ_i^+ , light refracts with angle of refraction $\theta_r^+ < \theta_i^+$.

Case II: Denser to rarer medium



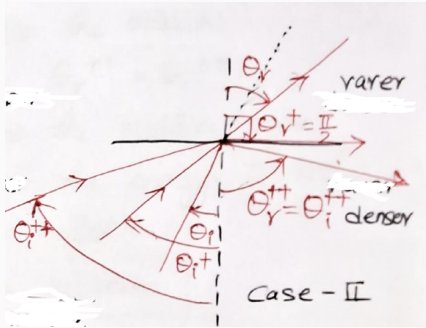
- Since light is slower in denser medium and faster in rarer medium, it bends away from the normal.
- Case (i) Angle of refraction is more than angle of incidence,

$$\theta_r > \theta_i.$$

- At a critical angle of incidence $\theta_i^+ = \theta_c$, the light grazes the interface. The angle of refraction is

$$\theta_r^+ = \frac{\pi}{2}$$

Total internal reflection

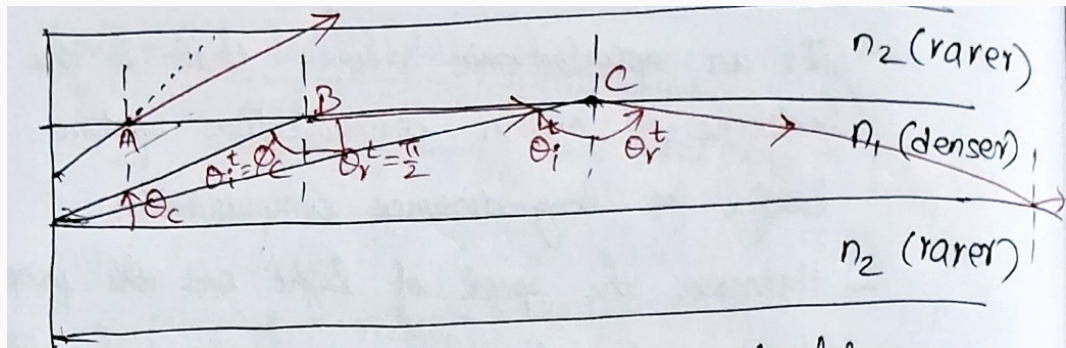


- Case (iii) For angle of incidence $\theta_i^{++} > \theta_c$, the light reflects back into the medium. The angle of reflection

$$\theta_r^{++} = \theta_i^{++}.$$

- Since the light enters the medium again with reflection and all the light energy is within the medium, this condition is called **Total Internal Reflection (TIR)**.
- TIR is required for light containment within the optical fiber. This possible if refractive index of core is greater than refractive index of cladding.

Optical fiber – conditions

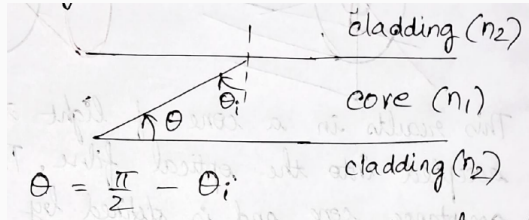


- Light is at one end of the optical fiber.
- Not all light can be contained within the light pipe (optical fiber).

Critical angle of propagation

Definition

Angle of propagation is defined as the angle between incident ray and core-cladding interface.



Acceptance angle

- Before the light enters the core medium, it has to encounter the interface between the core and the ambient ² medium.
- If the refractive index of ambient medium is n_3 , then for entering the core medium, the refractive index of ambient must be lesser than refractive index of core.
- For convenience, let us consider $n_3 = 1$. Therefore, the angle of incidence at the ambient-core interface needs to be less than the so called acceptance angle so that the refracted light is within the critical angle of propagation.
- At the acceptance angle of incidence θ_0 ,

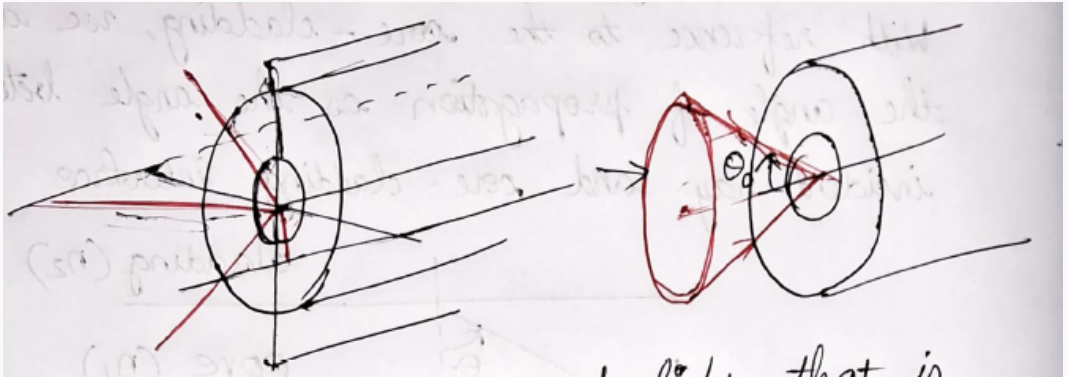
$$\frac{\sin \theta_0}{\sin \theta_0} = \frac{n_1}{n_3} = n_1$$

$$\theta_0 = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

²ambient means outside or commonly occurring, in our case it is usually air.

Acceptance cone

- We have limited our previous analysis to the "top" and "bottom" interface.
- Due to the cylindrical nature of optical fiber, the analysis is applicable to all planes passing through the axis.



This results in a cone of light that is accepted into the optical fiber. This is called acceptance cone and is defined by its acceptance angle.

- The industry has defined new quantities to quantify an optical fibre –
 1. numerical aperture
 2. fractional refractive index

Definition

Fractional refractive index is defined as relative difference in the refractive indices of core and cladding.

- The fractional refractive index is given by

$$\Delta = \frac{n_1 - n_2}{n_1}.$$

- In order to guide light effectively through a fiber, $\Delta \ll 1$. Typically, Δ is of the order of 0.01.

Numerical aperture

Definition

Numerical aperture is defined as the sine of the acceptance angle

$$\begin{aligned}\text{N.A.} &= \sin \theta_0, \\ &= \sin \left(\sin^{-1} \sqrt{n_1^2 - n_2^2} \right), \\ &= \sqrt{n_1^2 - n_2^2}, \\ &= \sqrt{(n_1 + n_2)(n_1 - n_2)}, \\ &= \sqrt{\left(\frac{n_1 + n_2}{2} \right) \left(\frac{n_1 - n_2}{n_1} \right)} \cdot 2n_1, \\ &\simeq \sqrt{n_1 \Delta 2n_1}, \\ &= n_1 \sqrt{2\Delta}.\end{aligned}$$

Optical fiber based communication

- Optical fiber is the backbone of modern communication system.

